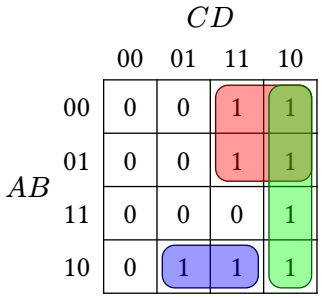
# the `k-mapper` package 1.3.0




```
#karnaugh(
  "4x4",
  labels: ($A$, $B$, $C$, $D$),
  manual-terms: (
    0, 0, 1, 1, 0, 0, 1, 1,
    0, 1, 1, 1, 0, 0, 1, 0
  ),
  implicants: ((3, 6), (2, 10), (9, 11)),
)
```

Diagram 1: Example implementation and code of a Karnaugh map with `k-mapper`.

## introduction

`k-mapper` is a Typst package for adding customizable Karnaugh maps of 2 by 2, 2 by 4, and 4 by 4 grid sizes to your Typst projects.

See the source code on the Github repository for the project here, and the changelog here.

## using `karnaugh()`

The main function of this package is the `karnaugh()` function, which allows you to create and customize all sizes of Karnaugh maps. See the following pages for function arguments.

### gray code position

The position of implicants in `k-mapper` are declared via *Gray code position*. This is similar to Karnaugh map packages in LaTeX.

The Gray code position of a cell is determined from the coordinates of that cell with respect to the binary axis labels.

The empty maps shown in Diagram 2 show each cell's Gray code position. Note that the Gray code position for a cell differs depending on the Karnaugh map's grid size.

Gray code position allows you to input minterms and maxterms using `manual-terms` simply by copying your truth table in that order.

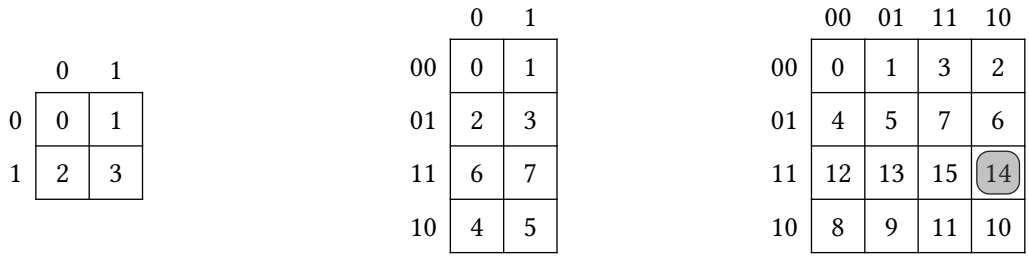



Diagram 2: Gray Code positions for three sizes of Karnaugh maps.

**function arguments**

| name | default | description | example values |
|---|---|---|---|
| **`grid-size`**<br>`int` | required | The size of the Karnaugh map's grid. This value can be only `"2x2"` (2 by 2), `"2x4"` (2 by 4), or `"4x4"` (4 by 4). Any other values will throw an error. | `2x2`<br>`2x4`<br>`4x4` |
| ~~**`x-label`**~~<br>`content` | `$$` | (Deprecated; use `labels`) ~~The label (usually a variable name) to go on the top (x-axis) of the Karnaugh map.~~ | `$A$`<br>`[foo]` |
| ~~**`y-label`**~~<br>`content` | `$$` | (Deprecated; use `labels`) ~~The label (usually a variable name) to go on the left (y-axis) of the Karnaugh map.~~ | `$B$`<br>`[bar]` |
| **`labels`**<br>`(string \| content)` | `()` | The labels (usually variable names) to be placed for their respective values. | `("A", "B", "C")`<br>`([bar],)` |
| **`style`**<br>`int` | `0` | The labeling style to be used. May be `0`, `1` or `2`; see map styles for examples. | `0` |
| ~~**`minterms`**~~<br>`(int)`<br>`none` | `none` | (Deprecated; use `manual-terms`) ~~The array of Gray code positions[1] where at that position is a minterm (`0`).~~<br>~~Mutually exclusive with `maxterms` and `manual-terms`.~~ | `(3, 4, 6)`<br>`(1, )` |
| ~~**`maxterms`**~~<br>`(int)`<br>`none` | `none` | (Deprecated; use `manual-terms`) ~~The array of Gray code~~ | `(0, 1, 2, 3, 5, 11, 12` |

[1] See p. 1.

| name | default | description | example values |
|---|---|---|---|
| | | ~~positions[1] where at that position is a maxterm (1).~~<br>~~Mutually exclusive with `minterms` and `manual-terms`.~~ | `(7, )` |
| **`manual-terms`**<br>`(content)`<br>`none` | `none` | The `array` of `content` in each cell in order of Gray-code position[1]. The length of this `array` *must* equal the `grid-size`.<br>Mutually exclusive with `minterms` and `maxterms`. | `// Grid-size 4`<br>`(0, "X", 1, 1)` |
| **`implicants`**<br>`((int, int), )` | `()` | An `array` where each element is an `array` of two `int`s, where each `int` is a Gray code position[1] corner of a *rectangular* implicant. | `((0, 3), (1, 1))`<br><br>`((0, 2), )` |
| **`horizontal-implicants`**<br>`((int, int), )` | `()` | An `array` where each element is an `array` of two `int`s, where each `int` is a Gray code position[1] corner of a *horizontal split* implicant — that is, one which wraps around the vertical edges of the Karnaugh map. | `// Grid-size 16`<br>`((0, 6), (8, 10))` |
| **`vertical-implicants`**<br>`((int, int), )` | `()` | An `array` where each element is an `array` of two `int`s, where each `int` is a Gray code position[1] corner of a *vertical split* implicant — that is, one which wraps around the horizontal edges of the Karnaugh map. | `// Grid-size 8`<br>`((0, 4), )`<br><br>`// Grid-size 16`<br>`((0, 9), (2, 10))` |

| name | default | description | example values |
|---|---|---|---|
| **`corner -implicants`** `bool` | `false` | A `bool` which indicates whether the Karnaugh map contains a `corner split` implicant — that is, one which wraps around both vertical and horizontal edges of the Karnaugh map. | `true` |
| **`cell-size`** `length` | `20pt` | The size of an individual cell in the Karnaugh map. | `1cm` |
| **`stroke-width`** `length` | `0.5pt` | The stroke width of the Karnaugh map grid. | `0.2pt` |
| **`colors`** `(color)` | array of:<br>red<br>green<br>blue<br>cyan<br>magenta<br>yellow | An array of RGBA `colors` to be used in displaying implicants. The first implicant uses the first `color` in the array, the second implicant the second color, etc. If there are more implicants than there are colors, each subsequent implicant will use the least recently used color (i.e. it wraps around).<br>By default, all `colors` in `colors` have alpha values of `100`. | `// Grayscale K-map`<br>`(rgb(`<br>`  200, 200, 200, 100`<br>`), )` |
| **`implicant-inset`** `length` | `2pt` | The inset of implicants within each cell. | `3pt` |
| **`edge-implicant -overflow`** `length` | `5pt` | How much *split implicants* (horizontal, vertical, corner) overflow the bounds of the grid. | `2mm` |
| **`implicant-radius`** `length` | `5pt` | The corner radius of implicants. | `3mm` |

| name | default | description | example values |
| --- | --- | --- | --- |
| **`implicant-stroke-transparentize`** `ratio` | `#-100%` | The ratio to transparentize the stroke color of implicants by. If set to `0%`, the stroke color of implicants are the same as the fill color, darked by the factor set in `implicant-stroke-darken` (`60%` by default). Negative values mean the stroke color becomes more opaque. | `-50%` |
| **`implicant-stroke-darken`** `ratio` | `60%` | The ratio to darken the stroke color of implicants by. | `100%` |
| **`implicant-stroke-width`** `length` | `0.5pt` | The stroke width of implicants. | `1pt` |

**map styles**

The `style` parameter can be used to dictate the variable labeling styles. The value can be `0`, `1`, or `2`. Courtesy of Aras14HD.

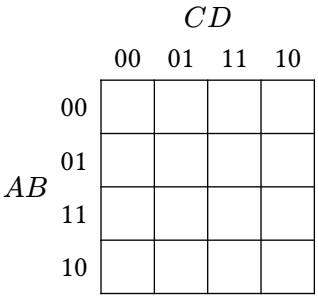



Diagram 3: Style `0`

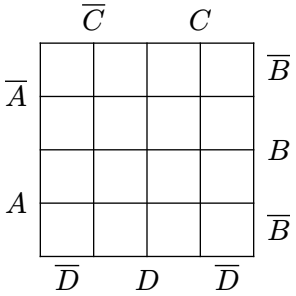



Diagram 4: Style `1`

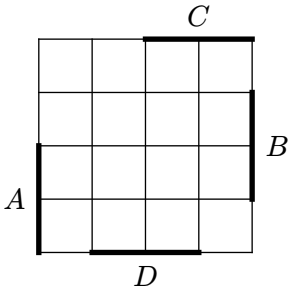


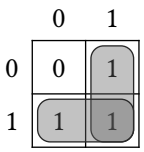

Diagram 5: Style `2`

**deprecation of `minterms` and `maxterms`**

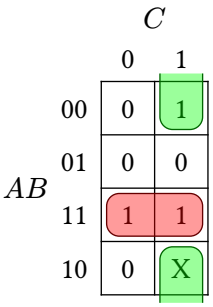
These two function parameters have been deprecated as I mixed up minterms and maxterms when initially writing this package. Rather than rename and risk breaking changes, I have decided to deprecate them and encourage the use of `manual-terms` instead.

Thank you to Moch. Nafkhan Alzamzami (`nafkhanzam`) on GitHub for pointing out this issue.

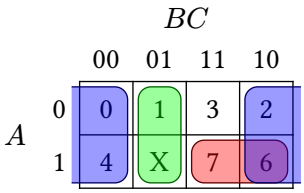
## examples



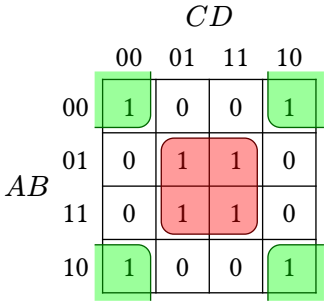

```
// Grayscale Karnaugh map
#karnaugh(
  "2x2",
  manual-terms: (0, 1, 1, 1),
  implicants: ((1, 3), (2, 3)),
  colors: (rgb(100, 100, 100, 100), ) // <-
)
```



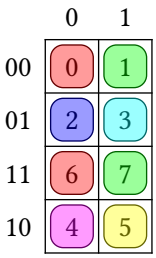

```
#karnaugh(
  "2x4",
  labels: ($A$, $B$, $C$),
  manual-terms: (0, 1, 0, 0, 0, "X", 1, 1),
  implicants: ((6, 7), ),
  vertical-implicants: ((1, 5), )
),
```



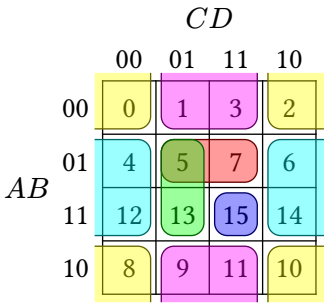

```
#karnaugh(
  "4x2",
  labels: ($A$, $B$, $C$),
  manual-terms: (0, 1, 2, 3, 4, "X", 6, 7),
  implicants: ((6, 7), (1, 5)),
  horizontal-implicants: ((2, 4), ),
),
```



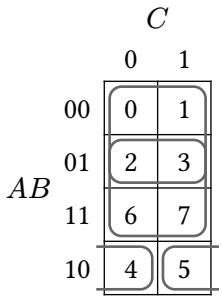

```
#karnaugh(
  "4x4",
  labels: ($A$, $B$, $C$, $D$),
  manual-terms: (
    1, 0, 1, 0, 0, 1, 0, 1,
    1, 0, 1, 0, 0, 1, 0, 1
  ),
  implicants: ((5, 15), ),
  corner-implicants: true
),
```




```
#karnaugh(
  "2x4",
  manual-terms: (0, 1, 2, 3, 4, 5, 6, 7),
  implicants: (
    (0, 0), (1, 1), (2, 2), (3, 3),
    (4, 4), (5, 5), (6, 6), (7, 7)
  )
)
```




```
#karnaugh(
  "4x4",
  labels: ($A$, $B$, $C$, $D$),
  manual-terms: (
    0, 1, 2, 3, 4, 5, 6, 7, 8,
    9, 10, 11, 12, 13, 14, 15
  ),
  implicants: ((5, 7), (5, 13), (15, 15)),
  vertical-implicants: ((1, 11), ),
  horizontal-implicants: ((4, 14), ),
  corner-implicants: true,
)
```




```
// No fill Karnaugh map
#karnaugh(
  "2x4",
  labels: ($A$, $B$, $C$),
  manual-terms: (0, 1, 2, 3, 4, 5, 6, 7),
  implicants: ((0, 3), (2, 7)),
  horizontal-implicants: ((4, 5), ),
  colors: (rgb(255, 255, 255, 0), ),
  implicant-stroke-width: 1pt
)
```